

Particulate Matter Monitoring & Health Effects

Issue 2026-08-10 | Entry window 10 August 2026 | 16 records in scope (3 pre-watch backfill)

16RECORDS IN SCOPE
(27 REJECTED)**10**SENSING / EXPOSURE
RECORDS (63%)**8**SUBTOPIC CLUSTERS
REPRESENTED**5**METADATA-ONLY
RECORDS**0**RATIO-SCALE EFFECT
ESTIMATES

Signal of the day

Two independent mobile-platform records land on the same day, and between them they define the measurement problem this watch keeps circling: dense mobile sampling produces the spatial detail that fixed networks cannot, and produces it in a form nobody has agreed how to calibrate.

Babu et al. (*Environ Monit Assess*) drove a size-resolved optical instrument over $\sim 2.4 \text{ km}^2$ of Madurai at a 6 s cadence on a 50 m grid and recovered a **bimodal** distribution with peaks at $16.77\text{--}32.37 \mu\text{m}$ and $0.275\text{--}0.53 \mu\text{m}$, local PM_{10} enhancement of $188.91 \pm 149.07 \mu\text{g}/\text{m}^3$ over background, and Getis-Ord Gi^* hotspots locked to traffic corridors. Bobbia et al. (*Data in Brief*, pre-watch backfill surfaced by the Amass connector) publish the complementary artefact: **10 mobile AtmoTrack units on Rouen buses at 10 s, 18 fixed AirSensEUR units on traffic lights at 1 min, and 6 reference stations at 15 min** over 10.5 months, assembled explicitly to support *multi-hop* calibration — mobile sensor against fixed sensor against reference — because most mobile units never see a reference site. Read together the implication is uncomfortable and specific: Babu's standard deviation is **79% of the mean enhancement**, and on a moving platform that dispersion is inseparable from uncalibrated instrument response, sampling-line losses and 6 s response time. The Rouen dataset exists because that inseparability is the field's open problem; the Madurai paper reports hotspots without addressing it. Neither is wrong. The gap between them is where the calibration-transfer literature of the past fortnight — Ginsburg's regional PurpleAir corrections, Zhivkov's distance-dependent uncertainty — now has to go: from static co-location to platforms that move.

What changed today

- ▶ **The connector layer was widened, and it changed what the day looks like.** This issue is the first run to sweep the full set of connected research surfaces — PubMed, Amass BiomedCore, Consensus, Scholar Gateway, Semantic Scholar, arXiv, medRxiv, Crossref, Exa and Nimble — alongside the committed harvester. The measurable result is **not** more same-day records: PubMed’s entry-date window is still the only leg that reliably resolves a single day. What the wider sweep produced was **three high-value records with pre-watch entry dates** (9 July, 29 June, 9 June) that no date-windowed query could ever have surfaced, and **seven duplicates** that `seen.json` caught on PMID or DOI. Both outcomes are worth having; neither substitutes for the entry-date spine.
- ▶ **Instrument-side particle metrics finally outnumber mass metrics.** Six of sixteen records report absorption coefficient, number size distribution, heavy-metal content or optical apportionment rather than a regulated mass concentration. The 9 August run log flagged that `pmclass()` had no bin for any of these and collapsed them all into “PM (unspecified)”. Four new classes were added in this issue. The fallback bin did not disappear — it still holds 4 records — but those four now genuinely have no stated particle metric, and 5 records that would have been buried there are separable as optical properties or composition/speciation.
- ▶ **Source apportionment moves from composition to optics.** Acton-Bond et al. apply *rolling* positive matrix factorisation to aerosol *optical* apportionment rather than to filter-based chemical speciation. If the approach holds, it puts source attribution within reach of an aethalometer rather than a laboratory, which is directly relevant to what a distributed network can be asked to report. The record is metadata-only, so this is a flag, not a finding.
- ▶ **The measurement-comparability question is being asked out loud.** Wang et al. title a paper “operational definitions, method comparability and environmental interpretation” for heavy metals in PM. That is the same critique this watch has been making of low-cost mass measurement, applied to a different quantity: the number you report is a property of your protocol before it is a property of the air.
- ▶ **Health-side records are all weak designs today, and one of them is instructive about why.** Chin et al. regress monthly counts of HS clinical encounters on monthly counts of days above $12\ \mu\text{g}/\text{m}^3$ and report $\beta = 0.57$ at lag 0 and $\beta = 0.62$ at lag 2, with lag 1 and lag 3 null. A non-monotonic lag structure across four adjacent lags in a single-centre monthly series is a pattern of multiple comparisons, not a biological latency.
- ▶ **No forest plot again — seventh consecutive issue.** Nothing in the corpus is ratio-scale with an interval. The two quantitative health estimates are a **−11.8 g** birth-weight difference per $10\ \mu\text{g}/\text{m}^3$ second-trimester $\text{PM}_{2.5}$ and a **+23.1%** preterm-birth risk increase, both quoted without CIs in a narrative review. `f5_forest.png` is intentionally absent.
- ▶ **Standing gap, now measured over 469 records:** still not one paper couples an instrument to a health endpoint. The MARTHA device flagged on 9 August (10.1016/j.jaerosci.2026.106882) remains without an abstract in Crossref or Europe PMC after 72 hours, so it cannot yet be counted.

Corpus analytics

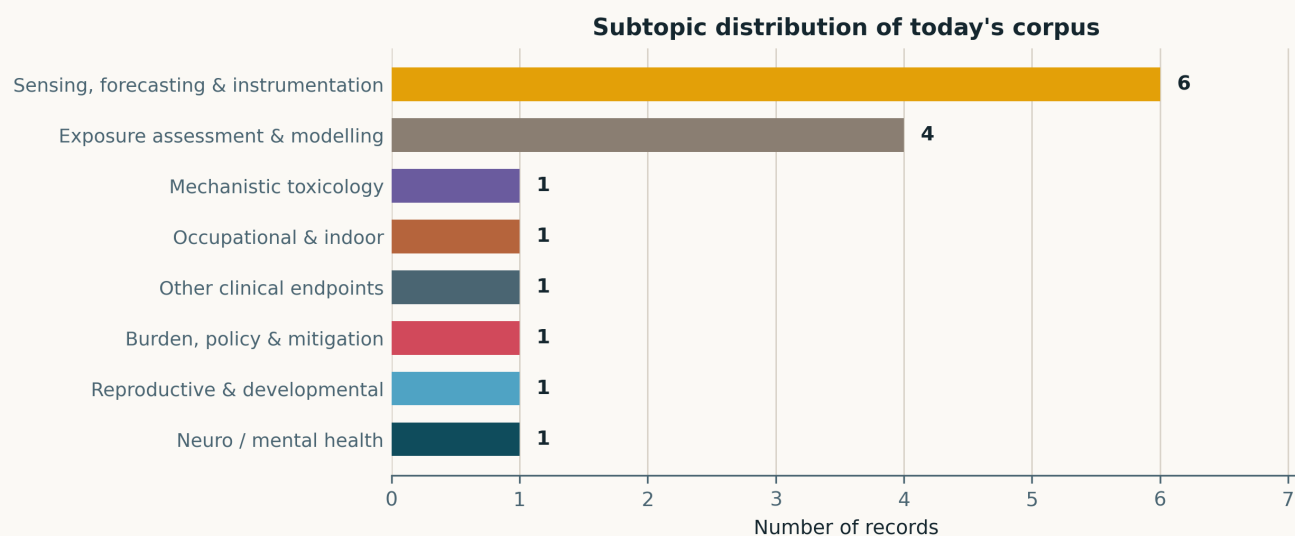


Fig. Subtopic assignment of the 16 in-scope records. Assignment is single-label by dominant contribution. Sensing/instrumentation (6) and exposure assessment (4) together take 63% of the issue; the six health-side subtopics carry one record each. The flat tail is the shape of a day whose health literature was thin and whose measurement literature was not.

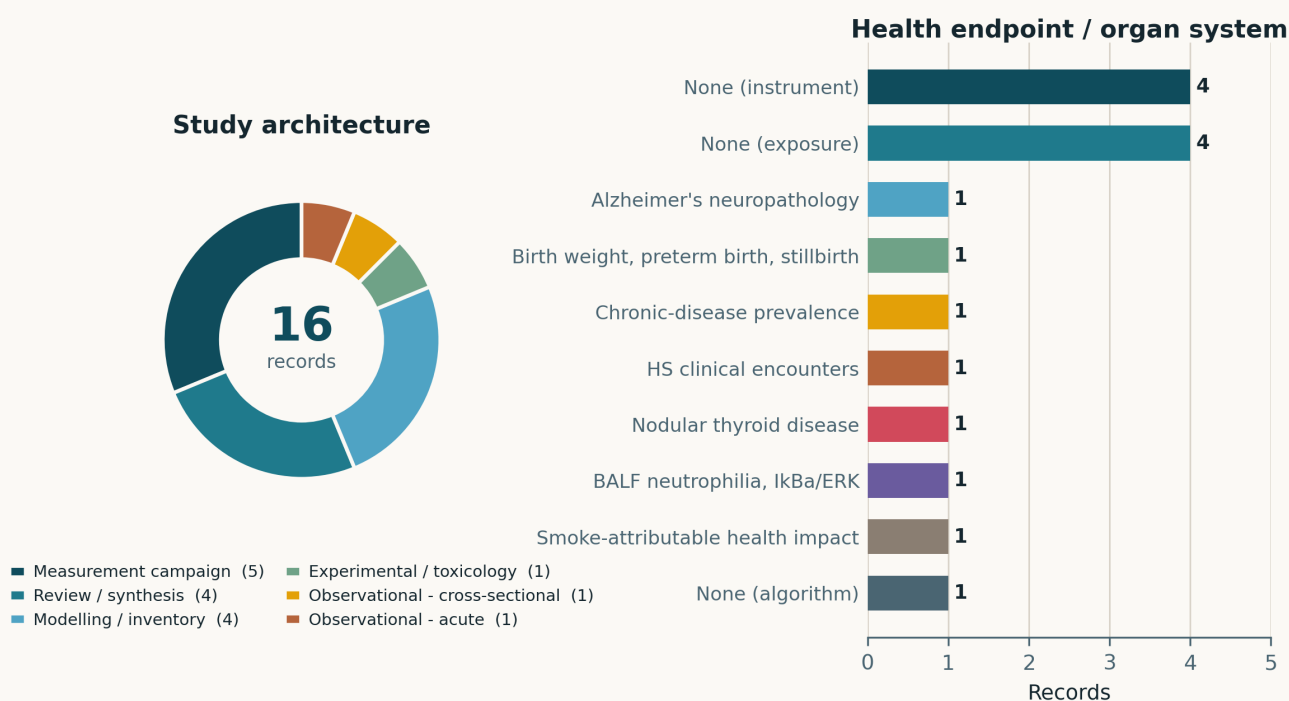


Fig. Study architecture (left) and endpoint class (right). Measurement campaigns lead at 5, with review/synthesis and modelling/inventory tied at 4 each and one record each in experimental/toxicology, observational cross-sectional and observational acute. On the right, 9 of 16 records carry no health endpoint at all — 4 instrument, 4 exposure-surface, 1 algorithm — and the remaining seven each report a different endpoint, so no health outcome appears twice. That is the arithmetic of a 63% measurement corpus, and it is why this issue has no forest plot.

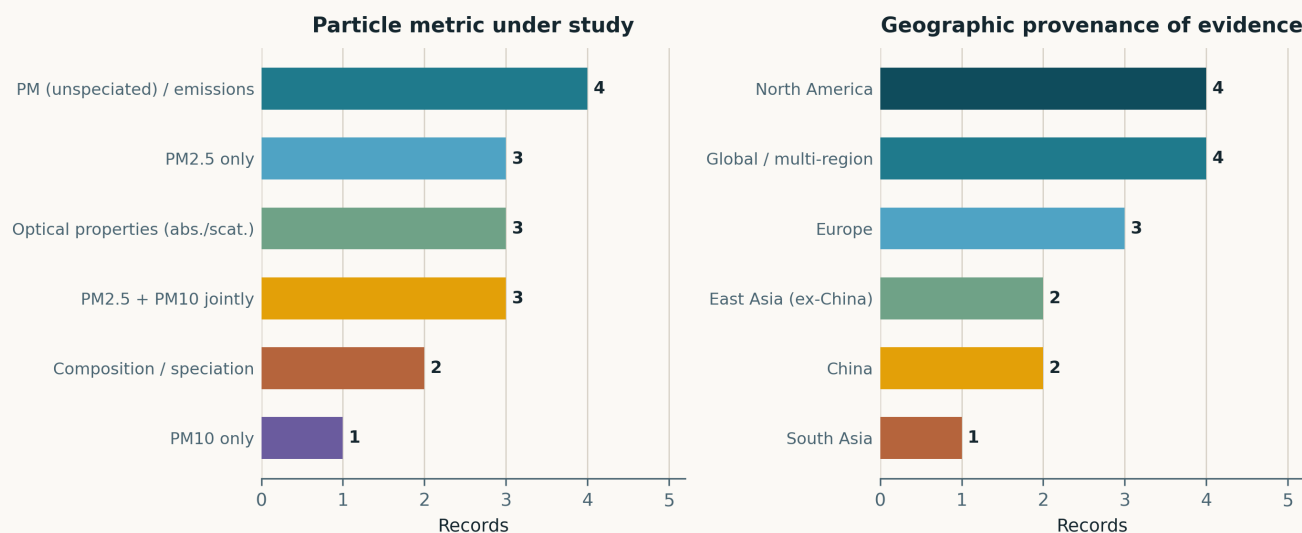


Fig. Particle metric (left) and geographic provenance (right). The left panel uses the particle-metric vocabulary extended in this issue. It has not eliminated the fallback: “PM (unspeciated) / emissions” is still the largest single bin at 4, but it now holds only records that genuinely have no stated particle metric (an unspecified-pollutant algorithm, unmeasured occupational dust, a qualitative pollution exposure and a multi-pollutant-class review) rather than absorbing every instrument-side quantity as it did through 9 August. **Optical properties** now carries 3 and **composition / speciation** 2, against 0 before the fix. On the right, North America and the global/multi-region bucket tie at 4; Europe has 3, China and East Asia 2 each, South Asia 1. The global/multi-region count includes **two** records that are honestly non-geographic (a method review and an algorithm paper) plus two genuinely multi-region reviews, so it should not be read as evidence of internationally pooled fieldwork.

Life-course windows addressed today (records may span several stages)

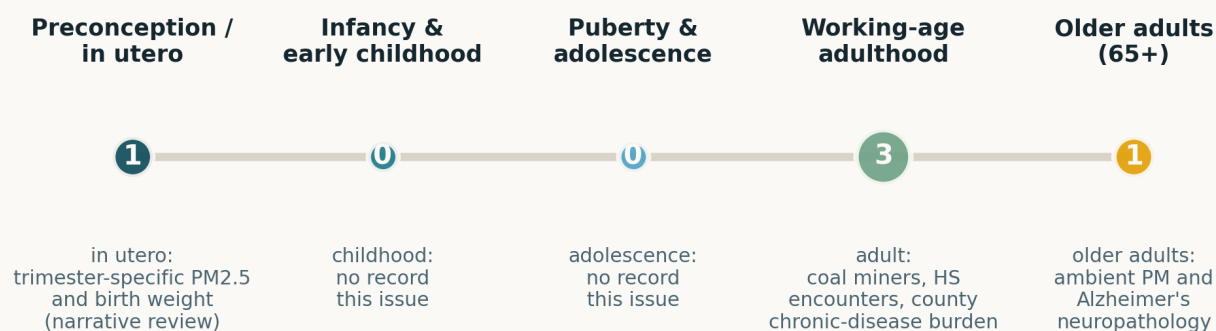


Fig. Life-course coverage. One in-utero record (trimester-specific PM_{2.5} and birth weight), three adult, one older-adult, and **nothing at all in childhood or adolescence** — a gap that has now persisted across most issues of this watch and that the 2–8 August weekly also recorded. Counts are of records, not participants.

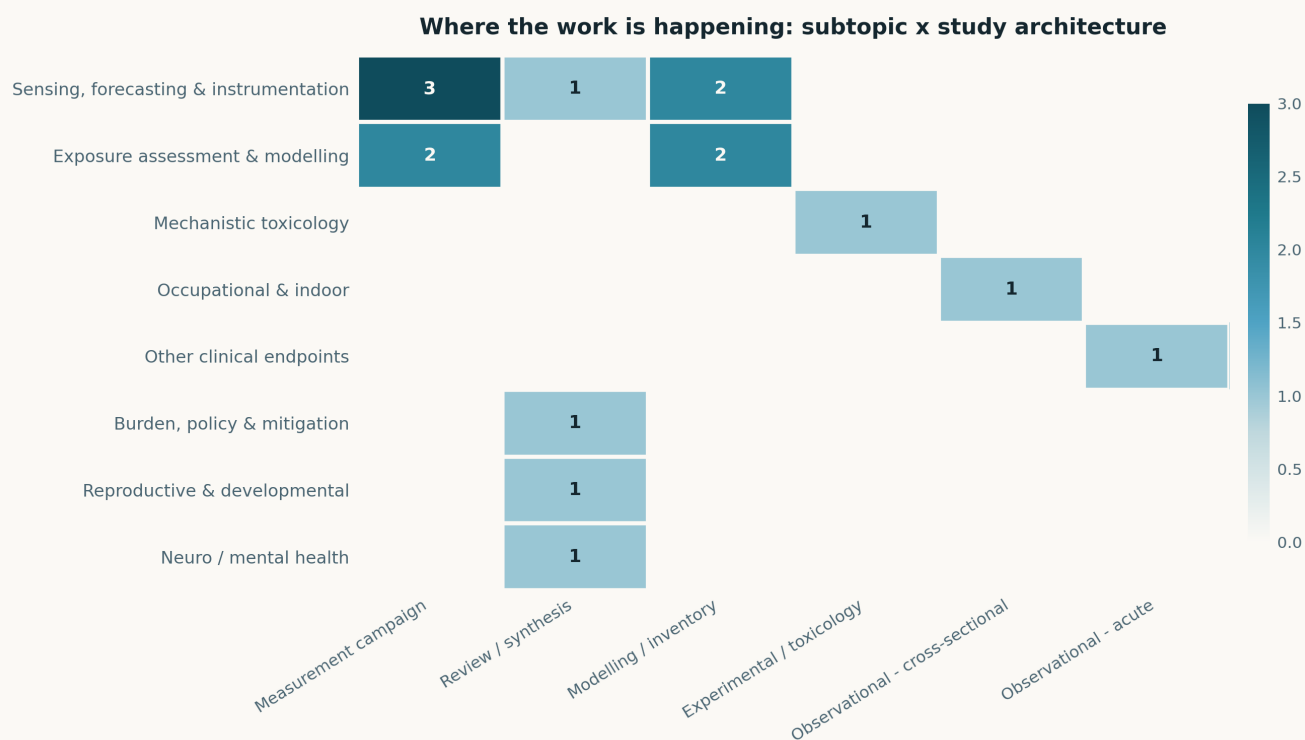


Fig. Subtopic $\times$ study architecture. The two blocks are disjoint, and that is the finding. Sensing and exposure occupy measurement campaigns (3 and 2) and modelling/inventory (2 and 2) and appear **nowhere** in the experimental or observational columns; the six health-side subtopics occupy review/synthesis (3) and one cell each in experimental/toxicology, observational cross-sectional and observational acute, and appear **nowhere** under measurement campaign. No record in this issue sits in both halves. Across all sixteen issues of this watch — **469 records** — that intersection has never been populated, which makes it structural rather than a sampling artefact of any one day.

Paper-level digest

Records are grouped by subtopic and ordered within group by first author. Tier A/B/C is a judgement about how much weight the design can carry, not about the journal. Records marked **METADATA ONLY** had no abstract in Crossref, Europe PMC, Semantic Scholar or Scholar Gateway at build time and are listed rather than summarised. Records marked **BACKFILL** have entry dates before this watch began on 26 July 2026 and were surfaced by a relevance-ranked connector, not by the date window.

Sensing, forecasting & instrumentation

6 records

Acton-Bond et al. 2026 Atmos Environ

Rolling positive matrix factorisation for aerosol optical apportionment

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because *rolling* PMF applied to optical rather than chemical data is a specific and consequential claim: conventional PMF source apportionment requires filter sampling and offline speciation, whereas an optical implementation with a rolling window could in principle run on an aethalometer or a multi-wavelength absorption photometer at deployment cadence. That would move source attribution from a laboratory capability to a network capability, which is the distinction that decides whether a distributed sensing programme can report anything beyond mass. Whether the paper supports that reading is unknown until an abstract appears.

doi: 10.1016/j.atmosenv.2026.122295

Akazezuwa et al. 2026 Environ Monit Assess

SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence

Proposes an “Intelligent Environmental Monitoring Framework” combining adaptive filtering, Isolation Forest outlier removal, min–max normalisation, multi-order neighbour feature fusion and a reinforcement-learning optimiser, reporting **97.1 % accuracy, 97.8 % F1, 97.6 % recall and 98.21 % precision**. Those figures cannot be interpreted. The abstract names no pollutant, no dataset, no sensor, no units, no sample size and no held-out split; “accuracy” is quoted for what is elsewhere described as predicting continuous pollutant concentrations, which is a category error unless the task was silently discretised. A four-metric cluster in the 97–98 % band on environmental time series with autocorrelation is the signature of evaluating on temporally adjacent data. Carried as tier C because the framework class (edge-side denoising plus learned correction) is squarely on this watch’s axis and the record should be visible in the register; the reported numbers should not be cited.

doi: 10.1007/s10661-026-15724-0

Babu et al. 2026 Environ Monit Assess

Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city

Mobile monitoring campaign in Madurai, Tamil Nadu, during the northeast-monsoon retreat: a vehicle-mounted size-resolved optical instrument sampling every **6 s** across **~2.4 km²** aggregated to a **50 m × 50 m** grid. Local enhancement above route background was largest for PM₁₀ at **188.91 ± 149.07 µg/m³**, attributed to road dust and construction; evening PM₁ (**42.10 µg/m³**) exceeded morning levels; the size distribution was **bimodal**, with coarse (16.77–32.37 µm) and submicron (0.275–0.53 µm) peaks. Getis-Ord Gi* identified persistent hotspots on traffic corridors and cold spots in restricted-access zones; SEM-EDX on 22 elements linked size to composition. The limitation that threatens the inference is that the reported dispersion (SD 79 % of the mean) is not decomposed: on a moving platform, true spatial variance, uncalibrated optical response, inlet losses and 6 s instrument response time are confounded, and no co-location against a reference monitor is described. Directly relevant to distributed sensing: this is the spatial resolution a mobile network buys, reported without the calibration chain that would make it transferable.

doi: 10.1007/s10661-026-15746-8

Bobbia et al. 2026 Data in Brief

Multi-source dataset of NO₂ and PM from mobile and fixed sensors in Rouen (France)

BACKFILL (PubMed entry 2026-07-09; surfaced by the Amass connector, not by the date window). Data descriptor for the TIGA “Smart Mobility for All” deployment in Rouen: **10 mobile AtmoTrack** units on buses at 10 s, **18 fixed AirSenseEUR** units on traffic lights at 1 min, **6 reference stations** at 15 min, plus 2 AtmoTrack units co-located beside a reference station, over **15 January–30 November 2025**. Mobile records carry bus speed, GPS position and onboard RH/T/pressure; fixed records add O_x, NO and CO. The stated purpose is explicitly **multi-hop calibration** — transferring a reference-derived correction through fixed sensors to mobile units that never see a reference site — and heterogeneous data fusion. This is the single most directly usable record of the issue for distributed low-cost PM sensing, because it is the rare case where the co-location topology needed to test calibration transfer is published rather than described. Limitation: it is a data descriptor, so it contains no calibration result, no error metric and no validation; the value is entirely in the availability of the raw structure.

doi: 10.1016/j.dib.2026.113025

Gilardoni et al. 2026 Atmos Chem Phys

First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps

First continuous **number size distribution and absorption coefficient** record at 1 h resolution from the Testa Grigia Observatory (3480 m a.s.l., Italian Alps), spanning **September 2021–May 2023**. Three aerosol population types are separated by transport pathway: a coarse population indicating long-range Saharan transport, a fine population tied to Po Valley and alpine-valley boundary-layer dynamics and broader continental flow, and low-number periods representing free-tropospheric background. The record is a baseline dataset rather than a hypothesis test, and the limitation is the one every single-site campaign carries — population attribution rests on back-trajectory and covariance reasoning, not on independent source measurement. Relevant to distributed sensing for a narrow but useful reason: a high-altitude site with characterised free- tropospheric periods gives an absolute-zero reference for optical instruments that low-altitude urban co-location cannot provide, and the absorption-coefficient record is one of the few instrument-side quantities published at deployment cadence.

doi: 10.5194/acp-26-11135-2026

Wang et al. 2026 Atmos Environ

Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because the title states the problem this watch treats as central, transposed to a different measurand: for heavy metals in PM, the reported concentration depends on the digestion protocol, the filter substrate, the size cut and whether total or bioaccessible fraction is meant — so “comparability” is a property of the method chain, not of the air. The same argument applies to low-cost optical mass measurement, where the reported µg/m³ is a function of assumed density, refractive index and hygroscopic growth. A systematic treatment of operational definitions on the metals side would be worth reading against the sensor-calibration literature.

doi: 10.1016/j.atmosenv.2026.122291

Exposure assessment & modelling

4 records

Fan et al. 2026 Atmos Environ*Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried because it is the first record in this watch to attach a health-impact estimate to the January 2025 Los Angeles fires, and because the physical premise is specific and testable: offshore Santa Ana flow reverses the usual coastal transport direction, so smoke-attributable exposure is not predicted by proximity to the burn perimeter. Satellite work on the same event (Li et al., *AGU Advances*, Feb 2026) reports spread rates to 3.7 km h^{-1} , residential-area fire intensities of $72\text{--}297 \text{ W m}^{-2}$ and a total-PM emission coefficient of $10.2\text{--}21.1 \text{ g MJ}^{-1}$, with 24 h mean $\text{PM}_{2.5}$ within 7.5 km reaching $300 \mu\text{g/m}^3$ — context, not this paper's result. The health-impact methodology is unknown until an abstract appears, and for a wildland–urban-interface fire burning structures the applicability of vegetation–fire concentration–response functions is exactly the assumption that needs checking.

doi: 10.1016/j.atmosenv.2026.122292

Gong et al. 2026 Res Sq (preprint)*Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas*

Constructs a **100 m** grid over the Seoul metropolitan area, kriges a continuous surface from **193 monitoring stations** for SO_2 , CO, NO_2 , O_3 , PM_{10} and $\text{PM}_{2.5}$ combined into a Comprehensive Air-quality Index, reduces 15 urban-form variables to four principal components (urban enclosure, low-density buildings, high-density buildings, development intensity) and fits geographically weighted regression. CAI rises from **88–92** in northeastern Seoul to **96–104** in the southwest. The headline model comparison — adjusted R^2 **0.9928** for GWR against **0.0182** for OLS — should be read as a warning rather than a result. A 54-fold jump in explained variance on a kriged surface means the GWR is largely reproducing the interpolation, since kriged values at 100 m between 193 stations are smooth by construction and locally varying coefficients will fit that smoothness almost exactly. No spatial cross-validation or effective-degrees-of-freedom correction is reported. Tier C. Carried because the urban-form-to-exposure framing is the right one for siting a sensor network, and because it is a clean illustration of how interpolated exposure surfaces manufacture model fit.

doi: 10.21203/rs.3.rs-10443285/v1

Hernandez et al. 2026 Atmos Environ*Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city*

METADATA ONLY — no abstract in Crossref, Europe PMC or Scholar Gateway. Listed, not summarised. Carried at tier C because atmospheric microplastic is a particle class that optical monitors detect but do not identify: fibres in particular violate the spherical-equivalent-diameter assumption underlying every optical particle counter, so a deposition record that quantifies their abundance as a function of urbanisation and humidity bears on the size and shape distribution a low-cost sensor is actually sampling. The humidity dependence is the part worth reading if an abstract appears, since RH is simultaneously the dominant artefact in optical PM measurement and, here, an exposure variable.

doi: 10.1016/j.atmosenv.2026.122294

Lin et al. 2026 Atmos Chem Phys*Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects*

Identifies an elevated aerosol layer embedded *within* the daytime mixed layer over the Hexi Corridor using intensive vertical-profile campaigns in **2010 and 2012** at Dunhuang and Minqin. Excluding dust events, elevated layers occurred at **0.6–2 km** on **82 %** of days at Dunhuang and **44 %** at Minqin, with the paper analysing formation mechanism, air-quality consequence and radiative effect. The relevance to surface monitoring is direct and under-appreciated: a concentration maximum aloft but inside the mixing layer breaks the well-mixed assumption that lets a surface monitor stand for a column, and it is precisely the configuration in which satellite AOD and ground $\text{PM}_{2.5}$ decouple — which is the failure mode of every AOD-to-surface-PM scaling relationship. Limitations: two campaigns fourteen years old, two sites, and an arid region whose boundary-layer behaviour does not generalise to the humid urban settings where most low-cost networks are deployed.

doi: 10.5194/acp-26-11111-2026

Mechanistic toxicology

1 record

Cho et al. 2026 J Microbiol Biotechnol*Lactacaseibacillus paracasei ATG-E1 alleviates particulate-matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites*

BALB/c mice challenged intranasally with PM_{10}D (PM_{10} plus diesel exhaust particles) on days 0, 3, 6 and 8, then treated orally with ATG-E1 or dexamethasone *after* inflammation was established — a therapeutic rather than prophylactic design, which is the methodological point in its favour. ATG-E1 reduced BALF inflammatory infiltrate and neutrophilia, lowered collagen deposition and pro-inflammatory mediators, and attenuated **$\text{I}\kappa\text{B}\alpha$ and ERK phosphorylation** while leaving JNK and p38 unchanged; caspase-1 and IL-1 α fell, lung RNA-seq showed broad downregulation of cytokine–cytokine-receptor signalling, and cecal 16S plus faecal metabolomics showed microbiome remodelling with increased branched-chain fatty acids and decreased branched-chain amino acids. Two limits: the microbiome and metabolite changes are correlative, with no germ-free, antibiotic or transfer arm to establish that the gut axis mediates the airway effect rather than co-occurring with it; and no effect size, n per group or variance is given in the abstract, so the magnitude is unassessable. All authors are employed by the company that owns the strain.

doi: 10.4014/jmb.2604.04029

Occupational & indoor exposure

1 record

Zhao et al. 2026 J Int Med Res*Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities*

Cross-sectional analysis of **1,708** coal miners examined at an occupational-disease hospital in Anhui in April 2021, combining general, laboratory and imaging data to predict nodular thyroid disease, using a proposed “Nonlinear Inverse Nearest Manifold Projection” model benchmarked against established risk models. The record is carried at tier C and should not be cited for a result: the abstract reports that the model “achieved high scores in both F1 and area under the precision–recall curve” and “significantly outperformed” comparators **without giving a single numeric value**, confidence interval, or comparator name. Beyond the missing numbers, dust exposure is never measured — occupation is the exposure proxy, with no mass concentration, no respirable fraction, no duration and no job-exposure matrix — so the paper cannot support any dose–response claim about particulate exposure and thyroid nodules. Included because occupational particulate exposure with thyroid endpoints is a thin literature and the register should show what is in it.

doi: 10.1177/03000605261473853

Reproductive & developmental

1 record

Hahad et al. 2026 J Environ Health Sci Eng*Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings*

BACKFILL (PubMed entry 2026-06-29; surfaced by the Amass connector). Narrative review synthesising epidemiological and mechanistic evidence linking prenatal air pollution to low birth weight, preterm birth and stillbirth, with searches across PubMed, Scopus and Web of Science. The two quoted estimates are a **−11.8 g** birth-weight difference and a **+23.1 %** increase in preterm-birth risk per **10 $\mu\text{g}/\text{m}^3$** second-trimester $\text{PM}_{2.5}$; mechanisms discussed are oxidative stress, inflammation, endocrine disruption, vascular dysfunction and epigenetic modification acting on placental morphology, perfusion and metabolic function. Both estimates are quoted **without confidence intervals or provenance to a specific pooled analysis**, which is why neither enters the forest plot — a narrative review is not a meta-analysis, and a point estimate lifted from one without its interval is not an effect size. The review is honest about what remains unresolved: disentangling individual pollutants, establishing causality, and identifying critical windows. Relevant to sensing because trimester-specific windows demand exposure resolution that residential-address annual means cannot deliver.

doi: 10.1007/s40201-026-00990-4

Neuro / mental health

1 record

Domingo 2026 Arch Toxicol*Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease*

BACKFILL (PubMed entry 2026-06-09; surfaced by the Amass connector). Mechanistic review drawing on human cohorts, animal models and *in vitro* neuronal/glial systems to link pollutant classes — ambient PM, heavy metals, pesticides, PFAS and micro/nanoplastics — to Alzheimer's-relevant neuropathology, framed against the 2024 *Lancet* Commission's inclusion of air pollution among 14 modifiable dementia risk factors and a projected **>150 million** cases by 2050. Shared pathways discussed are neuroinflammation, oxidative stress, blood-brain-barrier disruption, tau-kinase dysregulation, epigenetic reprogramming and gut-brain dysbiosis, with particular weight on reports of AD-type neuropathology in children and young adults in heavily polluted urban environments. That evidence is the review's most striking claim and also its weakest link: it rests substantially on one research programme's Mexico City autopsy series, where selection into autopsy, APOE distribution and co-exposure to metals and traffic noise are not separable from PM. Read as an argument for exposure reduction as a dementia-prevention lever, it is reasonable; read as evidence of a PM-specific causal pathway, it is not yet that.

doi: 10.1007/s00204-026-04469-x

Burden, policy & mitigation

1 record

Vartanyan et al. 2026 Cureus*Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework*

Comparative descriptive analysis of public-health data for **2018–2021** showing Fresno County, California, exceeding state prevalence and mortality for obesity, hypertension, diabetes, asthma, coronary heart disease and stroke, attributed to poverty, low educational attainment, provider shortages, food insecurity and severe air and pesticide pollution, with disproportionate impact on racial and ethnic minority populations. Tier C, and the reason is structural rather than a criticism of execution: this is an ecological comparison of county aggregates in which air pollution is one of several correlated exposures, none measured at individual level, so no exposure-specific inference is available and none is claimed. Carried because the San Joaquin Valley is among the most $\text{PM}_{2.5}$ -burdened airsheds in the United States and because the paper's underlying argument — that monitoring density and exposure burden are inversely related in marginalised communities — is the same one Siu et al. quantified for eastern Canada in the 5 August issue, here asserted rather than measured.

doi: 10.7759/cureus.112418

Other clinical endpoints

1 record

Chin et al. 2026 Res Sq (preprint)*Impact of air quality and $\text{PM}_{2.5}$ on hidradenitis suppurativa activity in the Sacramento region*

Retrospective analysis of **10,431** daily hidradenitis suppurativa encounters at UC Davis Health, **April 2015–March 2022**, regressing monthly encounter counts on the monthly count of days exceeding **12 $\mu\text{g}/\text{m}^3$** $\text{PM}_{2.5}$ from the EPA Air Quality System, with lags to three months. Positive associations at **lag 0** ($\beta = 0.5658$, $P = 0.022$) and **lag 2** ($\beta = 0.6230$, $P = 0.015$); lag 1 and lag 3 null. The lag structure is the problem. A significant effect at 0 and 2 months with nothing at 1 month has no biological reading, and four lags tested without adjustment on a monthly series of ~ 84 points is a multiple-comparisons pattern. Three further limits: healthcare encounters measure care-seeking as much as disease activity and are strongly seasonal; a regional monitor assigned to a referral population is a substantial exposure misclassification; and no autocorrelation correction or seasonal adjustment is described, which for monthly counts over seven years will inflate significance on its own. Preprint, single centre, not peer reviewed.

doi: 10.21203/rs.3.rs-10570042/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Domingo	Aging in a highly polluted world: challenges and solutions to prevent Alzheimer's disease	<i>Arch Toxicol</i>	Mechanistic review	Ambient PM + metals, PFAS, micro/nanoplastics	Multi-region
Reproductive & developmental					
Hahad et al.	Air pollution and adverse birth outcomes: a narrative review of epidemiological and mechanistic findings	<i>J Environ Health Sci Eng</i>	Narrative review	PM _{2.5} , trimester-specific	Multi-region
Mechanistic toxicology					
Cho et al.	Lactacisbacillus paracasei ATG-E1 alleviates particulate matter-induced airway inflammation in association with altered inflammatory signalling, gut microbiota and faecal metabolites	<i>J Microbiol Biotechnol</i>	Animal model	PM ₁₀ D (PM ₁₀ + diesel exhaust particles)	East Asia
Occupational & indoor					
Zhao et al.	Occupational dust exposure and thyroid nodule screening: a predictive modelling approach for early detection in coal mining communities	<i>J Int Med Res</i>	Cross-sectional + prediction model	Occupational coal dust (unmeasured)	China
Sensing, forecasting & instrumentation					
Acton-Bond et al.	Rolling positive matrix factorisation for aerosol optical apportionment	<i>Atmos Environ</i>	Source-apportionment method	Absorption coefficient / optical apportionment	Europe
Akazuwa et al.	SFFO-RL intelligent environmental monitoring: smart sensing networks with artificial intelligence	<i>Environ Monit Assess</i>	Algorithm development	Unspecified pollutant concentration	Not stated
Babu et al.	Capturing the invisible: hyperlocal variability of size-resolved particulate matter in a smart city	<i>Environ Monit Assess</i>	Mobile monitoring campaign	PM ₁ , PM _{2.5} , PM ₁₀ + size distribution	India
Bobbia et al.	Multi-source dataset of NO ₂ and PM from mobile and fixed sensors in Rouen	<i>Data in Brief</i>	Network deployment / data descriptor	PM ₁₀ , PM _{2.5} , NO ₂ + RH/T/P	France
Gilardoni et al.	First continuous aerosol measurements at Testa Grigia at 3480 m a.s.l.: aerosol populations and dust transport dynamics in the southern European Alps	<i>Atmos Chem Phys</i>	Measurement campaign	Number size distribution + absorption coefficient, 1 h	Europe
Wang et al.	Measuring heavy metals in atmospheric particulate matter: operational definitions, method comparability and environmental interpretation	<i>Atmos Environ</i>	Method review	Heavy-metal content of PM	Not stated
Exposure assessment & modelling					
Fan et al.	Santa Ana winds control wildfire smoke transport and related health impacts in California during January 2025	<i>Atmos Environ</i>	Chemical transport modelling	Wildfire-sourced PM _{2.5}	North America
Gong et al.	Regional analysis of urban physical factors influencing air pollution: evidence from Seoul's living areas	<i>Res Sq (preprint)</i>	Spatial regression (GWR)	Composite AQ index (incl. PM _{2.5} , PM ₁₀)	East Asia
Hernandez et al.	Urbanization and humidity impact atmospheric deposition of anthropogenic microparticles (including microplastics) in a humid subtropical city	<i>Atmos Environ</i>	Deposition sampling	Anthropogenic microparticle deposition flux	North America
Lin et al.	Elevated aerosol layers within the daytime mixed layer over ecologically sensitive areas of northwest China: diurnal variation, formation mechanisms and regional effects	<i>Atmos Chem Phys</i>	Measurement campaign	Aerosol vertical profile, 0.6-2 km layer	China
Burden, policy & mitigation					
Vartanyan et al.	Structural inequities and chronic disease in Fresno County: a narrative review and multi-sectoral framework	<i>Cureus</i>	Narrative review	Air + pesticide pollution (qualitative)	North America
Other clinical endpoints					
Chin et al.	Impact of air quality and PM _{2.5} on hidradenitis suppurativa activity in the Sacramento region	<i>Res Sq (preprint)</i>	Retrospective time series	Days above 12 µg/m ³ PM _{2.5} (monthly count)	North America

Provenance and method

Entry window **2026-08-10 to 2026-08-10**, contiguous with the 9 August issue; no gap. This is the first issue to run the **full connected connector set**, and the per-leg yield is recorded here because the negative results are as informative as the positive ones. **PubMed** connector, three independent queries against [Date - Entry]: health axis 2, instrumentation axis 1, broad MeSH sweep 11 — 13 distinct records, **6 carried**. Local harvester: PubMed health 5, PubMed sensing 0, Europe PMC CREATION_DATE 3 (**2 carried**), Crossref by ISSN across 17 journals 12 (**7 carried**). **Amass BiomedCore** returned 10 relevance-ranked records: 4 were duplicates caught by `seen.json`, 3 were declined pre-watch backfills, **3 carried as flagged backfill** with entry dates 9 July, 29 June and 9 June. **Consensus** returned 10, **0 carried** — 4 were duplicates already shipped on 26 July, 5 August and 8 August, the rest resolved outside the window; this is the fourth consecutive run confirming that Consensus is a relevance index and cannot see a single day. **Scholar Gateway semanticSearch** could not resolve any of the five Elsevier abstracts: its corpus is Wiley full text with a stated last update of **May 2026**, so it is a passage retriever for older literature, not a recency or cross-publisher leg. **arXiv** 60 records, **1** published on or after 26 July and off-topic. **medRxiv**, **Semantic Scholar** (rate-limited), **Exa** and **Nimble** returned nothing inside the window. **OpenAlex** HTTP 429, **eighth consecutive run**. **ClinicalTrials.gov** returned 311 matching trials but none registered or updated on 10 August; `trials.json` unchanged, and the trial watch is surfaced in the weekly rollup, not here. **27 records rejected**, each with a reason in `state/rejected.jsonl`. **Five carried records have no abstract** in Crossref, Europe PMC, Semantic Scholar or Scholar Gateway — all five are Elsevier *Atmospheric Environment* DOIs, the same publisher-side gap first logged on 8 August, now persisting past 72 hours. They are marked `METADATA ONLY`. Per the 9 August finding, the abstract re-check has been moved to a weekly sweep. **No forest plot is generated for this issue**: the corpus contains no ratio-scale estimate with an interval, so `f5_forest.png` is intentionally absent rather than missing. Two `plots.py` defects were closed in this issue. `pmclass()` gained four instrument-side particle-metric classes (optical properties; size-resolved / number distribution; composition / speciation; composite index) plus hyphen normalisation, closing the defect logged on 9 August. Separately, `design_group` fell through for **12 of 16** records on the first build — the ninth recurrence of that defect, and the worst — which rendered `f2`'s architecture panel as a single "Other / mixed" block; all twelve labels are now mapped and the figure was regenerated. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record. Content derived from PubMed, Crossref, Europe PMC and Amass metadata; abstracts remain the copyright of their respective publishers.